

Zhihan Zhou

Personal Website / LinkedIn / Email / Google Scholar

I am a Research Scientist at **NVIDIA**, working on **AI for Genomics** and **Large Language Models**.

EDUCATION

Northwestern University

- ✧ Doctor of Philosophy, Computer Science (GPA: **4.0/4.0**) 09/2020 - 03/2025
- ✧ Master of Science, Computer Science (GPA: **4.0/4.0**) 09/2018 - 06/2020

Zhejiang University

- ✧ Bachelor of Science, Mathematics and Applied Mathematics 09/2014 - 06/2018

WORK EXPERIENCE

Research Scientist @ NVIDIA.

03/2025 - present

- ✧ Develop novel AI algorithms for secondary genome analyses.

Research Intern @ NVIDIA

05/2024 - 09/2024

- ✧ Build long-context sequence model for simultaneous germline and somatic structural variants calling.

Research Intern @ Snap

06/2023 - 10/2023

- ✧ Develop foundation models to generate general-purpose user embedding for millions of Snap users. Conduct comprehensive research in data representation, model design, training objective, and evaluation pipelines.
- ✧ Leads to two academical papers and a system that is being developing and deploying to the product.

Applied Scientist Intern @ Amazon Web Services.

09/2021 - 12/2021

- ✧ Conduct an independent research project to improve few-shot dialogue understanding by large-scale contrastive conversational model pre-training. Findings and models are published in NAACL 2022.

SELECTED PUBLICATIONS

- ✧ **DNABERT: Pre-trained Bidirectional Encoder Representation from Transformers for DNA Language Modeling in Genome**
Yanyong Ji, Zhihan Zhou*, Han Liu, Ramana V Davuluri (*: joint first author) ([Over 2 million downloads](#))*
Bioinformatics, 2021
- ✧ **DNABERT-2: Efficient Foundation Model and Benchmark for Multi-Species Genome**
Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana V Davuluri, Han Liu ([Over 3 million downloads](#))
ICLR 2024
- ✧ **GenomeOcean: An Efficient Genome Foundation Model Trained on Large-Scale Metagenomic Assemblies**
Zhihan Zhou, Robert Riley, Satria Kautsar, Weimin Wu, Rob Egan, Steven Hofmeyr, Shira Goldhaber-Gordon, Mutian Yu, Harrison Ho, Fengchen Liu, Feng Chen, Rachael Morgan-Kiss, Lizhen Shi, Han Liu, Zhong Wang ([Over 3 million downloads](#))
Preprint, 2024
- ✧ **DNABERT-S: Pioneering Species Differentiation with Species-Aware DNA Embeddings**
Zhihan Zhou, Weimin Wu*, Harrison Ho, Lizhen Shi, Ramana V Davuluri, Zhong Wang, Han Liu (*: joint first author)*
ISMB 2025

AWARDS

- ✧ Terminal-year Ph.D. Fellowship @ Northwestern University

ACADEMIC SERVICES

- ✧ Serve as Reviewer for: *NeurIPS*, *ICML*, *ICLR*, *ACL*, *EMNLP*, *NAACL*, *AISTATS*, etc.
- ✧ Serve as Teaching Assistant for:
CS 348: Introduction to Artificial Intelligence; CS 397: Seminar in Statistical Language Modeling; CS 496-6: Advanced Topics in Deep Learning, CS 496-9: Statistical Machine learning, CS 497: Deep Learning for Natural Language Processing, Stats 359: Data Science Project

OTHER PUBLICATIONS

- ✧ **USE: Dynamic User Modeling with Stateful Sequence Models**
*Zhihan Zhou**, *Qixiang Fang**, *Leonardo Neves*, *Francesco Barbieri*, *Yozen Liu*, *Han Liu*, *Maarten W. Bos*, *Ron Dotsch* (*: joint first author)
Preprint, 2024
- ✧ **General-Purpose User Modeling with Behavioral Logs: A Case Study with Snapchat**
*Qixiang Fang**, *Zhihan Zhou**, *Leonardo Neves*, *Francesco Barbieri*, *Yozen Liu*, *Dong Nguyen*, *Daniel L. Oberski*, *Maarten W. Bos*, *Ron Dotsch* (*: joint first author)
SIGIR 2024
- ✧ **Efficient Action Robust Reinforcement Learning with Probabilistic Policy Execution Uncertainty**
Guanlin Liu, *Zhihan Zhou*, *Han Liu*, *Lifeng Lai*
Transaction on Machine Learning Research, 2024
- ✧ **KGML-xDTD: A Knowledge Graph-based Machine Learning Framework for Drug Treatment Prediction and Mechanism Description**
Chunyu Ma, *Zhihan Zhou*, *Han Liu*, *David Koslicki*
GigaScience, 2023
- ✧ **Learning Dialogue Representations from Consecutive Utterances**
Zhihan Zhou, *Dejiao Zhang*, *Wei Xiao*, *Nicholas Dingwall*, *Xiaofei Ma*, *Andrew Arnold*, *Bing Xiang*
NAACL 2022
- ✧ **Trade the Event: Corporate Events Detection for News-based Event-driven Trading**
Zhihan Zhou, *Liqian Ma*, *Han Liu*
ACL 2021, *Findings*
- ✧ **Few-shot Slot Tagging with Collapsed Dependency Transfer and Label-enhanced Task-adaptive Projection Network**
Yutai Hou, *Wanxiang Che*, *Yongkui Lai*, *Zhihan Zhou*, *Yijia Liu*, *Han Liu*, *Ting Liu*
ACL 2020
- ✧ **Joint Speaker Diarization and Recognition Using Convolutional and Recurrent Neural Networks**
Zhihan Zhou, *Yichi Zhang*, *Zhiyao Duan*
ICASSP 2018
- ✧ **ELLPMDA: ensemble learning and link prediction for miRNA-disease association prediction**
*Xing Chen**, *Zhihan Zhou**, *Yan Zhao* (*: joint first author)
RNA Biology, 2018

SKILLS

- ✧ Programming languages: Python, Java, R, HTML, CSS
- ✧ Skills: Deep Learning (Pytorch, HuggingFace, DeepSpeed, etc.), Cloud Computing (Kubernetes, Docker, AWS, GCP, etc.), Bioinformatics (pbsim, minimap, BLAST, etc.)